A 28-nm 16-Kb SRAM Computing-in-Memory Macro With Dual Bitline Computing for Boolean Logic Operation and BWN MAC Operation

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***Abstract*—By integrating computation directly within memory structures, computing-in-memory (CIM) effectively mitigates the von Neumann bottleneck, offering significant improvements in energy efficiency and system throughput. This letter presents a 16-kb static random access memory (SRAM) CIM macro in 28-nm complementary metal–oxide–semiconductor (CMOS), featuring a novel dual bitline computing cell (DBCC) architecture. The design supports three operational modes: conventional memory access, Boolean logic operations, and binary weight network (BWN) multiply-and-accumulate (MAC) computations. The DBCC architecture addresses key critical limitations of prior works by: 1) storing both positive and negative weights within a single 6T-SRAM cell, improving array utilization by 2× compared to segregated storage schemes; 2) achieving consistent discharge rates for both polarities through symmetric pull-down paths, enhancing calculation accuracy [INL <3 least significant bit (LSB) across corners] and process variation tolerance; and 3) enabling differential voltage-based Boolean logic operations without external reference voltage generation, thereby reducing peripheral circuitry. The design features a compact 2:1 multiplexer (2:1 MUX)-based pulsewidth modulator (PWM) for 5-b signed input encoding and demonstrates 55.8–204.8-TOPS/W energy efficiency in 28-nm CMOS. Experimental results on a 16-kb array show stable operation across PVT variations while supporting both memory functions and in situ computation for BWNs. System-level evaluation demonstrates its practicality, achieving competitive inference accuracies of 97.59% on MNIST and 89.08% on CIFAR-10.**

***Index Terms*—Binary weight network (BWN), computing-in-memory (CIM), dual bitline computing cell (DBCC), process variation tolerance, signed multiply-and-accumulate (MAC) operation.**

I. Introduction

THE rapid advancement of artificial intelligence (AI) technology has driven a surge in demand for computational power [1]. Under the traditional von Neumann architecture, data transfer consumes 100 times more energy than computation itself, severely restricting AI computing efficiency [2]. Computing-in-memory (CIM) directly processes data in memory, eliminating the von Neumann bottleneck caused by data transfer between the processor and memory [3], [4], [5]. Static random access memory (SRAM) CIM has excellent energy efficiency and throughput, which can drive revolutionary breakthroughs in AI application prospects [6]. However, when deploying convolutional neural networks (CNNs), high-precision parameter storage decreases computational efficiency, separated positive/negative weight designs reduce hardware utilization, and charging/discharging rate variations with process sensitivity compromise computational stability, collectively limiting high-performance mobile deep CNNs (DCNNs) applications [7], [8], [9]. Recent SRAM CIM innovations address partial challenges, such as 8T-SRAM designs that achieve ultralow dynamic power and enhanced reliability [10], as well as split dual-port architectures that support area-efficient throughput doubling [11]. However, these approaches still face tradeoffs in signed computation support, array utilization, and peripheral complexity when scaling to multimode operations.

First, regarding computation architecture, many designs utilize the SRAM array primarily as a digital weight memory, while performing critical analog summation or digital accumulation in separated peripheral modules such as column analog-to-digital converters (ADCs) [12] or shared auxiliary computation cells [9], [13]. This separation constrains parallelism and energy efficiency. For example, 8T-SRAM CIM macros using column ADCs for neural network tasks face limited parallelism due to the peripheral quantization dependency [10]. Likewise, 4T1T SRAM-based engines implement shared peripheral circuits for both feedforward and backpropagation but still isolate computation from the storage arrays [14]. Second, for compact in-cell signed computation, some approaches store positive and negative weights in separate cells [15], [16], [17], which can result in k-bit weights occupying 2k SRAM cells and reducing array utilization. Others integrate computation within the cell but face linearity issues, such as relying on a single, asymmetric charge/discharge path [18], or employing inherently mismatched pMOS and nMOS paths for positive and negative results, which necessitates active compensation circuits [19]. In addition, charge-domain

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computing schemes, while compact, also must carefully manage the nonlinearity and mismatch introduced by the analog charge transfer paths [7], [15]. Notably, voltage-mode designs like the 8T-SRAM CIM [11] achieve high linearity via voltage division but lack native signed computation support. Third, for in-memory Boolean logic, conventional schemes are either limited in operand count [20], [21] or require a stable external reference voltage (V_{ref}) for sensing [22], [23], increasing peripheral complexity. Even dual-read SOT-MRAM CIM architectures that support Boolean operations rely on external sense amplifiers (SAs) with reference voltages [24], while an 8T compute-SRAM bitcell for digital CIM simplifies logic implementation but does not extend to multioperand signed computation [25].

To address these issues, we propose a novel dual bitline computing cell (DBCC)-based SRAM CIM macro.

1) To move beyond the compute-at-periphery paradigm, the DBCC embeds parallel analog accumulation within the array by using the complementary bitlines (PBL/NBL) as intrinsic summation nodes, avoiding separate peripheral compute modules, which is distinct from designs that rely on column ADCs [25] or shared auxiliary cells [14].

2) To achieve compact and robust signed computation, it stores positive and negative weights in a single 6T-SRAM cell, doubling array utilization compared to [15] and [17], while retaining 6T-SRAM's simplicity. Crucially, it performs signed computation using symmetric, nMOS-only pull-down discharge paths for both polarities. This design overcomes the asymmetry and matching issues of prior asymmetric-path designs [18], [19] and the intrinsic nonlinearity challenges in charge-domain approaches [7], [15], ensuring high linearity and process tolerance without active compensation. These advantages are not fully realized in voltage-mode designs [11] or charge-sharing architectures [25].

3) To simplify in-memory logic, the architecture natively supports reference-free, multioperand Boolean logic by exploiting the differential PBL/NBL voltage, eliminating the external V_{ref} needed in [22] and [23] and dual-read MRAM CIMs [24]. In addition, a compact pulsewidth modulation circuit will not cause any overhead burden. This macro supports multiplication and accumulation operations with signed 5-b inputs and 5-b outputs, which can directly serve binary weight networks (BWNs). It performs excellently in terms of energy efficiency and has achieved reliable inference accuracy on benchmark datasets. Compared to full precision CNNs, this scheme adopts binary weights to compress storage requirements while maintaining model accuracy. It can be optimized for deployment needs in resource-constrained scenarios [26], [27] such as mobile devices.

II. SRAM CIM Macro Architecture and Multimode Operations

Fig. 1(a) shows the proposed SRAM CIM macro architecture based on DBCCs, which is centered around a local computing bank (LCB) module and includes modules such as a precharge circuit, decoding circuit, input driver circuit, mode switching circuit, and timing control circuit. The architecture integrates a 16-kb storage capacity of $64 \times 32 \times 8$.

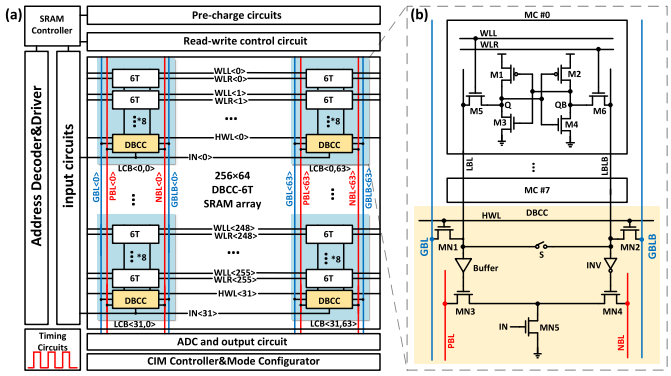


Fig. 1. Proposed (a) SRAM CIM macro architecture and (b) LCB circuit structure.

Fig. 1(b) shows the LCB circuit structure, which consists of eight 6T-SRAM cells with separated wordlines (WLs) and one DBCC, containing a total of 64×32 DBCCs. Among them, a DBCC consists of five transistors MN1–MN5, an inverter INV, a buffer, and a complementary metal–oxide–semiconductor (CMOS) transmission gate that functions as a switch S, which is jointly controlled by the global computation enable signal and the horizontal WL (HWL) to conduct only during Boolean logic and multiply-and-accumulate (MAC) operation modes. Also, the DBCC uses global bitlines (GBL/GBLB) as write bitlines, positive discharge line (PBL), and negative discharge line (NBL) as read and calculate bitlines. The dual WLs (WLL/WLR) and HWL control the conduction state of the 6T-SRAM cell and DBCC (6T-DBCC). The macro supports three working modes: SRAM conventional memory access, Boolean logic operation, and MAC operation. The specific functional implementation will be explained in detail in Sections II-A–II-C of this section.

A. SRAM Holding/Writing/Reading

Fig. 2 shows the basic principle of our proposed SRAM conventional memory access mode for 6T-DBCC. As shown in Fig. 2(a), when performing the holding operation, the WLL and WLR keep a low level, the latch structure composed of M1–M4 locks the data of the storage nodes Q/QB. In Fig. 2(b), when the 6T-DBCC performs a write operation to store 0 in the selected 6T-SRAM cell, the GBL is first discharged to ground, while GBLB is charged to the supply voltage VDD. Subsequently, the HWL is activated, allowing the voltage levels on GBL and GBLB to propagate to the corresponding local bitlines (LBL/LBLB) through transistors MN1 and MN2. The WLL and WLR of the 6T-SRAM are then activated, pulling down the originally stored 1 at node Q to a low level and pulling up node QB to a high level, thereby successfully writing and latching 0.

For the read operation shown in Fig. 2(c), prior to reading, GBL and GBLB are precharged to VDD. Taking reading 0 from the selected 6T-SRAM cell as an example, when HWL is activated, the voltage levels on GBL and GBLB are transmitted to corresponding LBL and LBLB through transistors MN1 and MN2, respectively. With WLL and WLR activated, the

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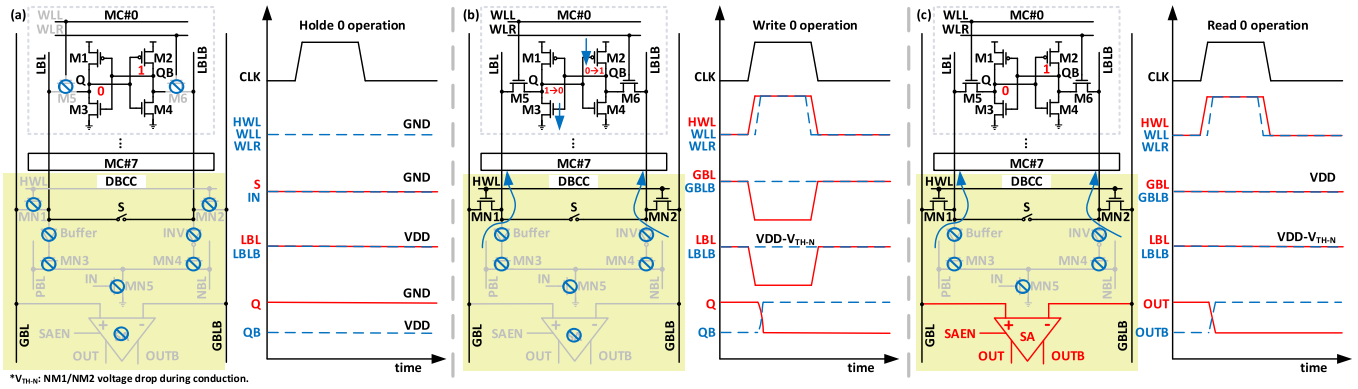


Fig. 2. Principles of SRAM conventional memory access mode for 6T-DBCC (a) holding, (b) writing, and (c) reading.

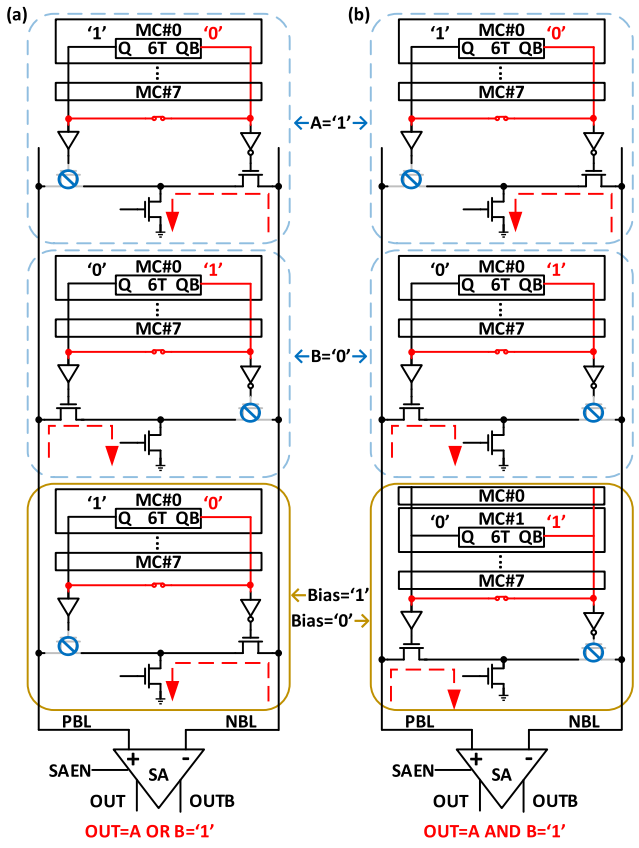


Fig. 3. Schematic of (a) OR operation and (b) AND operation of Boolean logic operation based on the DBCC architecture.

voltage on GBL gets pulled down, while GBLB maintains its precharged voltage. Finally, the voltage difference between GBL and GBLB is detected by the SA to read out 0.

B. Boolean Logic Operation Mode

When performing Boolean logic operations with 6T [28] and 8T [23] cell architectures, the bitline voltage decision relies on external V_{ref}. This single-ended detection mechanism is susceptible to process variations. Particularly during multirow collaborative computing, WL pulsewidth distortion induces bitline discharge gradient variations, where fixed V_{ref} settings cause quantization errors. Notably, even the improved 8T scheme proposed by Agrawal et al. [23] still requires V_{ref} for multirow decisions. Our proposed DBCC scheme innovatively achieves endogenous V_{ref} generation through real-time differential bitline voltage comparison, which not only reduces peripheral circuit complexity but also effectively mitigates V_{ref} drift risks caused by process fluctuations. More importantly, it eliminates the need for the dedicated, area-consuming analog voltage reference generator (e.g., bandgap reference) that is indispensable in conventional $V_{\text{ref-based}}$ schemes, thereby achieving a tangible reduction in overall peripheral overhead. By controlling redundant bias cells, multioperand OR and AND operations can be directly performed in the array. The result is read out by the SA, detecting the voltage difference between the bitlines PBL and NBL, without the need for the V_{ref} generation circuit required by traditional Boolean circuits. Fig. 3 takes the OR and AND operations of two operands A and B as an example, where A is set to 1 (Q = 1 and QB = 0) and B is set to 0 (Q = 0 and QB = 1).

TABLE I
OR OPERATION TRUTH TABLE OF TWO OPERANDS

A	B	Bias	V_{PBL}	V_{NBL}	OUT
0	0	1	VDD-2ΔV	VDD-ΔV	0
0	1	1	VDD-ΔV	VDD-2ΔV	1
1	0	1	VDD-ΔV	VDD-2ΔV	1
1	1	1	VDD	VDD-3ΔV	1

1) OR Operations: As shown in Fig. 3(a), when the data stored in the first-row 6T cell of the biased LCB are 1 (i.e., Q = 1 and QB = 0), the SA outputs the OR result of operations A and B. The computational process is as follows. After precharging the PBL and NBL to VDD, the WLRs of A and B are activated concurrently with the application of an input pulse (IN = VDD and pulsewidth = 1ΔT). At this time, A storing 1 and the bias cell (which consistently stores a fixed 1) discharge NBL by 1ΔV (totaling 2ΔV), while B storing 0 discharges PBL by 1ΔV. Consequently, the voltage on the PBL becomes VDD −1ΔV, while the voltage on the NBL becomes

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TABLE II
AND OPERATION TRUTH TABLE OF TWO OPERANDS

A	B	Bias	V_{PBL}	V_{NBL}	OUT
0	0	0	VDD-3ΔV	VDD	0
0	1	0	VDD-2ΔV	VDD-ΔV	0
1	0	0	VDD-2ΔV	VDD-ΔV	0
1	1	0	VDD-ΔV	VDD-2ΔV	1

TABLE III
OR/AND OPERATION TRUTH TABLE OF THREE OPERANDS

A	B	C	Offset row (OR/AND)	V_{PBL}	V_{NBL}	OUT (OR/AND)
0	0	0	1/0	VDD-3ΔV/ VDD-5ΔV	VDD-2ΔV/ VDD	0/0
0	0	1	1/0	VDD-2ΔV/ VDD-4ΔV	VDD-3ΔV/ VDD-ΔV	1/0
0	1	1	1/0	VDD-ΔV/ VDD-3ΔV	VDD-4ΔV/ VDD-2ΔV	1/0
1	1	1	1/0	VDD/ VDD-2ΔV	VDD-5ΔV/ VDD-3ΔV	1/1

VDD $-2\Delta V$. The SA detects this positive voltage difference (PBL voltage > NBL voltage) and thus outputs an OR logic result of 1. The corresponding operation truth table is shown in Table I.

2) AND Operations: As shown in Fig. 3(b), when the data stored in the first-row 6T cell of the biased LCB are 0 (i.e., Q = 0 and QB = 1), the SA outputs the AND result of operations A and B. The computational process is as follows. After precharging the PBL and NBL to VDD, the WLRs of A and B are activated concurrently with the application of an input pulse (IN = VDD and pulsewidth = $1\Delta T$). At this time, B storing 0 and the bias cell (which consistently stores a fixed 0) discharge PBL by $1\Delta V$ (totaling $2\Delta V$), while A storing 1 discharges NBL by $1\Delta V$. Consequently, the voltage on the PBL becomes VDD $-2\Delta V$, while the voltage on the NBL becomes VDD $-1\Delta V$. The SA detects this positive voltage difference (PBL voltage < NBL voltage) and thus outputs an AND logic result of 0. The corresponding operation truth table is shown in Table II.

Based on the above Boolean logic operation principle, for the case of N operands: during the calculation phase, the WLRs of the operand cell and bias row cell need to be set high, the switch S of the corresponding calculation cell needs to be closed, and IN needs to be set high. At this point, the input pulsewidth of the bias row calculation cell should be set to $(N-1)\ \Delta T$, while the calculation cell corresponding to the operand maintains the ΔT pulsewidth (ΔT is the reference pulsewidth for generating $1\Delta V$ bitline discharge). Tables III and IV, respectively, show the truth tables for OR and AND operations when the bias row pulsewidth is set to $(3-1)\ \Delta T$ for three operands and $(4-1)\ \Delta T$ for four operands.

TABLE IV
OR/AND OPERATION TRUTH TABLE OF FOUR OPERANDS

A	B	C	D	Offset row (OR/AND)	V_{PBL}	V_{NBL}	OUT (OR/ AND)
0	0	0	0	1/0	VDD-4ΔV/ VDD-7ΔV	VDD-3ΔV/ VDD	0/0
0	0	0	1	1/0	VDD-3ΔV/ VDD-6ΔV	VDD-4ΔV/ VDD-ΔV	1/0
0	0	1	1	1/0	VDD-2ΔV/ VDD-5ΔV	VDD-5ΔV/ VDD-2ΔV	1/0
0	1	1	1	1/0	VDD-ΔV/ VDD-4ΔV	VDD-6ΔV/ VDD-3ΔV	1/0
1	1	1	1	1/0	VDD/ VDD-3ΔV	VDD-7ΔV/ VDD-4ΔV	1/1

C. MAC Operation Mode

To overcome the bottleneck of the von Neumann architecture, BWN technology, based on single-bit quantization, replaces traditional multiplication with bit operations, significantly improving energy efficiency. Prior SRAM-based signed computation schemes, however, face inherent challenges in balancing accuracy and robustness. The scheme proposed by Kim et al. [18] uses a split WL 6T-SRAM cell and ties the bitlines to form a shared summation node (V_{SUM}). While area-efficient, it relies on a single, asymmetric charge/discharge path for both polarities, where the slower pMOS pull-up limits the charging rate and compromises linearity. An improved 10T design [19] employs dedicated but asymmetric computation paths: a pMOS-dominated path for positive results and an nMOS-dominated path for negative results. This approach directly handles signed weights but introduces a fundamental mismatch between pMOS and nMOS characteristics, making it sensitive to process variations and often necessitating active compensation circuits.

In contrast, the 6T-DBCC architecture proposed in this work adopts PBL/NBL dual discharge paths with a key innovation: using symmetric, nMOS-only pull-down paths for both positive and negative operations. This design offers several distinct advantages: 1) it eliminates the pMOS–nMOS mismatch inherent in [19], leading to better inherent matching; 2) it avoids the charge/discharge asymmetry of the shared path in [18]; and 3) it leverages the higher transconductance and lower sensitivity to threshold voltage variations of nMOS transistors compared to pMOS pull-up paths. Furthermore, the low-impedance discharge path to ground helps mitigate the impact of parasitic capacitance changes and isolates power supply noise, collectively enhancing computational stability and linearity. Fig. 4 shows the MAC operation process of the proposed 6T-DBCC architecture with 5-b input 1-b sign bit and 4-b data bit) and binary weights.

In the multiplication operation mode, the symbol encoding and weight storage encoding of the input signal in the 6T-DBCC architecture are both implemented through level states. As shown in Fig. 4(a), weight encoding is defined by the level combination of the 6T-SRAM storage node (Q/QB). When Q = 1 and QB = 0, the weight is + 1, and vice versa, the

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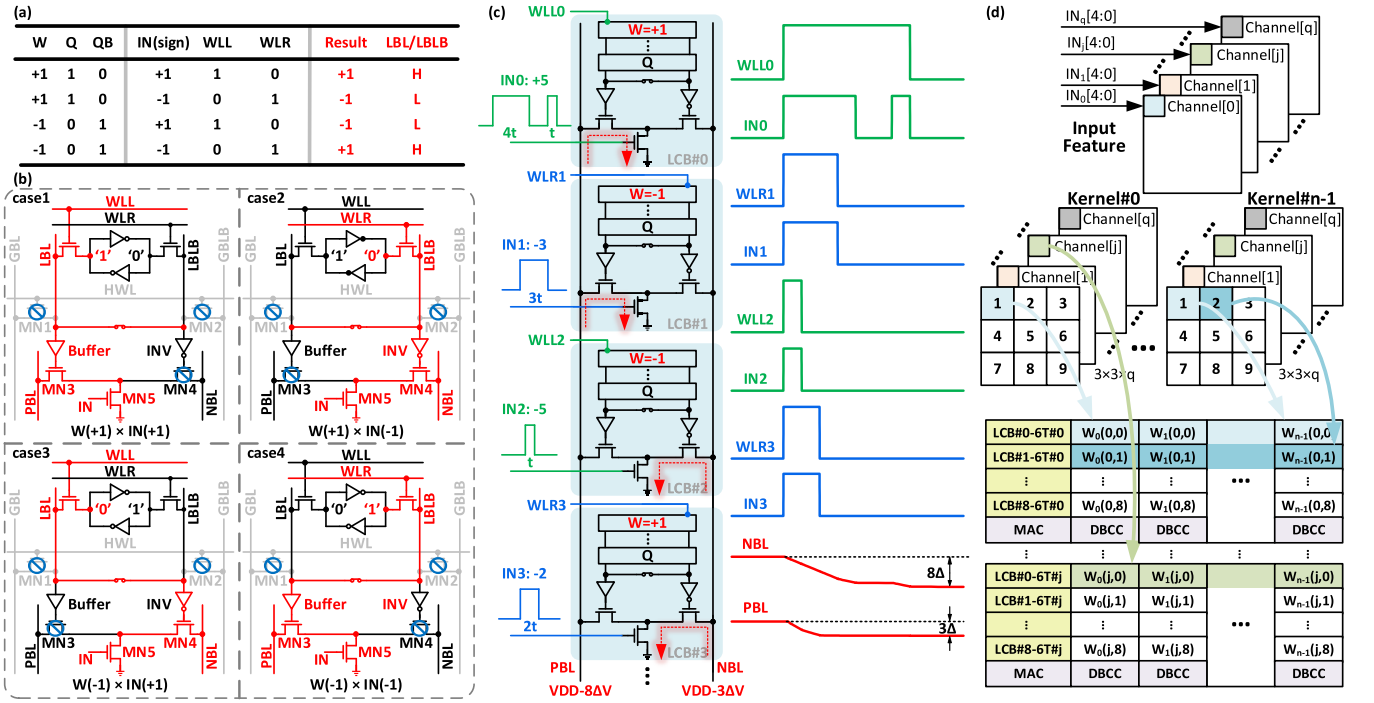


Fig. 4. (a) Signed encoding of input/weight/output, (b) multiplication principle, (c) MAC operation, and (d) BWN mapping in 6T-DBCC architecture.

weight is −1. The sign bit of a 5-b input signal is represented by a differential WL pair (WLL/WLR). When WLL = 1 and WLR = 0, the input is positive, and vice versa, the input is negative. Hardware friendly encoding of weights and input signal sign bits through level polarity matching can provide clear electrical state criteria for subsequent multiplication operations. The sign of the multiplication result is reflected in the LBL/LBLB level state, where LBL or LBLB is high; and the result is positive, and vice versa, the result is negative. The specific calculation is shown in Fig. 4(b). The switch S of 6T-DBCC is closed, and PBL/NBL is precharged to VDD. Taking weight + 1 as an example, the following conditions hold.

1) Case 1 (Multiplying Positive Input by + 1 Weight): Activate WLL, store data Q = 1/QB = 0 to pull LBL high, LBL signal drives MN3 after passing through the buffer. MN3 and input-driven MN5 form a conduction path to the pull-down PBL.

2) Case 2 (Multiplying Negative Input by + 1 Weight): Activate WLR, store data QB = 1/Q = 0 to pull LBLB low, LBLB signal drives MN4 after passing through the INV. MN4 and input-driven MN5 form a conduction path to the pull-down NBL.

3) Case 3 (Multiplying Positive Input by −1 Weight): Activate WLL, store data Q = 0/QB = 1 to pull LBL low, LBL signal drives MN4 after passing through the INV. MN4 and input-driven MN5 form a conduction path to the pull-down NBL.

4) Case 4 (Multiplying Negative Input by −1 Weight): Activate WLR, store data QB = 0/Q = 1 to pull LBLB high, LBLB signal drives MN3 after passing through the buffer. MN3 and input-driven MN5 form a conduction path to the pull-down PBL.

Fig. 4(c) shows a series of MAC mode operation examples. The 4-b data bit "abcd" ($a \sim d \in \{0, 1\}$) in the 5-b input signal is converted into a pulsewidth signal by a pulsewidth modulator (PWM) circuit, and its duration $M\Delta T$ is determined by the value $M = 8a + 4b + 2c + d$ (a total of 16 pulsewidth combinations, where ΔT corresponds to the pulsewidth of "0001"). This signal is input to the MN5 gate of the DBCC while accessing the binary weights stored in the 6T-SRAM through WLL/WLR (HWL disabled). The result of multiplying the weights by the input pulsewidth is mapped to the PBL/NBL voltage changes. The BL/BLB driven by the inverter and buffer generates fast voltage swings, controlling the on/off of MN3/MN4. During the whole column operation, the 6T-SRAM in multiple LCBs in the same row is synchronously activated. The voltage difference between PBL/NBL is read by the ADC, and the multibit multiplication and accumulation result is output.

Fig. 4(d) constructs a mapping model for BWN and 6T-DBCC, where W_n(j, i) represents the weight of BWN, consisting of n convolution kernels, $q + 1$ channels, and (j, i) represents the jth channel and ith number of convolution kernels. W_n(j, i) of the same convolution kernel is stored in the 6T-SRAM of the corresponding row in the same column but different LCBs. In MAC operation, each LCB calculates the product P_n(j, i) of the 5-b input and 1-b weight in (1). Within a single cycle, the MAC results of 32 DBCCs are accumulated into PBL and NBL, generating the partial and pMAC in (2). After completing all array calculations in T cycles, all parts are summed to obtain the final MAC value in (3), which is

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read and output by the ADC

$$P_n(j,i) = IN_j[4:0] \times W_n(j,i) \tag{1}$$

$$pMAC = \sum_{j=1}^{32} P_n(j,i) = \sum_{j=1}^{32} P_n(j,i)\, IN_j[4:0] \times W_n(j,i) \tag{2}$$

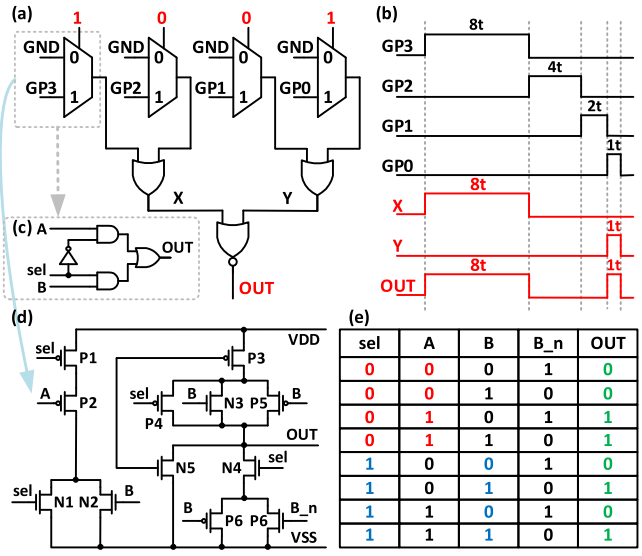
$$MAC = \sum_{t=1}^{T} pMAC = \sum_{t=1}^{T} pMAC \sum_{j=1}^{32} P_n(j,i). \tag{3}$$

III. Small-Area 2:1 Multiplexer in PWN Circuit

We use the PWM circuit in Fig. 5(a) to convert the 4-b data bits in the signed 5-b input signal into corresponding binary data. This configuration only needs to generate four reference pulsewidths to achieve any 4-b input pulsewidth signal. Fig. 5(b) shows the schematic of "1001" input signal pulsewidth generation. The PWM circuit requires four 2:1 MUXes. Implementing a 2:1 MUX with the combinational logic gate shown in Fig. 5(c) would consume $4 \times 20 = 80$ transistors. To minimize area overhead, we designed a compact 2:1 MUX in Fig. 5(d) utilizing only 12 transistors, achieving an 8T reduction compared to the conventional combinational logic design. This 2:1 MUX design uses sel as the enable signal, with A and B serving as data inputs. Based on the circuit schematic in Fig. 5(d) and the logic truth table in Fig. 5(e), it can be concluded that when sel = 1, the A terminal data are output, and when sel = 0, the B terminal data are output. Taking A = 0 and B = 1 as an example, the following conditions hold.

1) When sel = 0, VDD controls N5 to conduct and P3 to turn off through P1 and P2, VSS conducts through N5 tube, and outputs A = 0.

2) When sel = 1, VSS controls P3 to conduct and N5 to turn off through N1. At this point, B = 1 causes B_n = 0, N3/P5 transmission gates are conductive, P6/N6 transmission gates are closed, and VDD outputs B = 1 through the P3 transistor and N3/P5 transmission gates.



sel	A	B	B_n	OUT
0	0	0	1	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	1	0
1	1	1	0	1

Fig. 5. Input PWM circuit (a) schematic and (b) "1001" pulse generation illustration. (c) Combinatorial logic 2:1 MUX schematic. Small-area 2:1 MUX (d) schematic and (e) truth table.

IV. Evaluation Results

The proposed 16-kb SRAM CIM macro with DBCC was simulated using the 28-nm CMOS process.

A. Storage Mode Performance Simulation

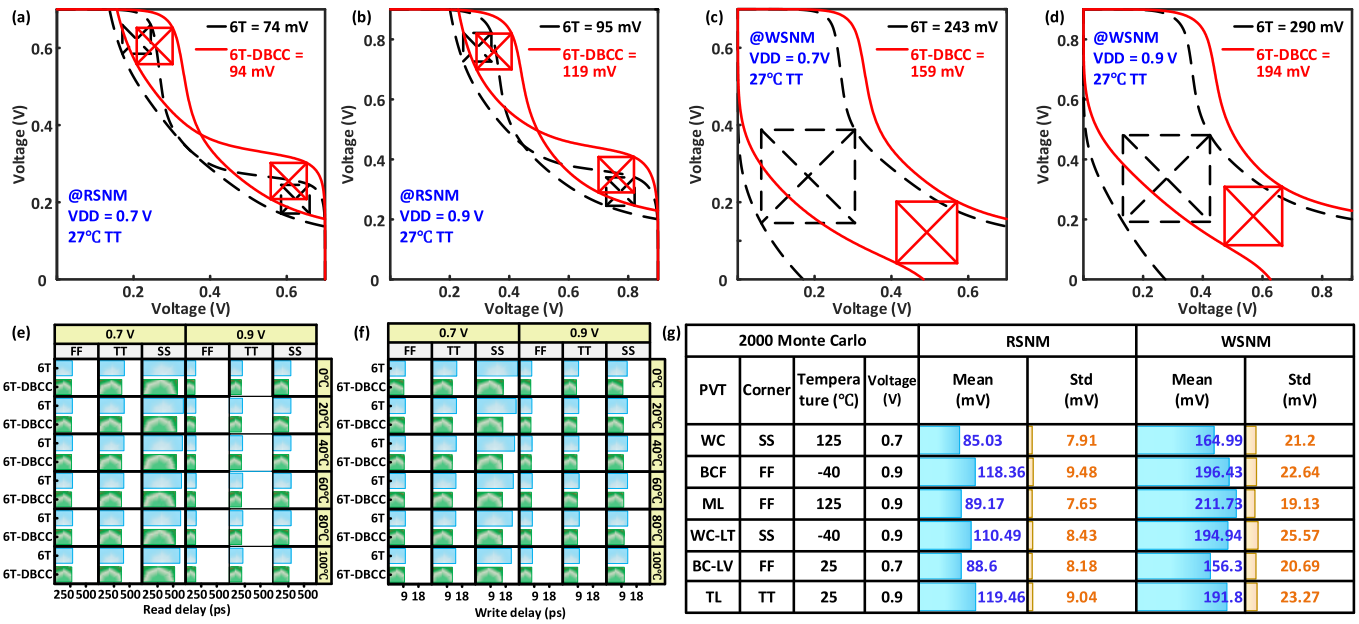
The stability of the SRAM cell in the conventional memory access mode can be evaluated through static noise margin (SNM). Fig. 6(a)–(d) compares the cell read SNM/write SNM (RSNM/WSNM) of 6T-SRAM and the proposed 6T-DBCC architecture under typical (TL) conditions (TT, 27 °C). The 6T-DBCC architecture accurately reads the weight values of storage cells in the selected LCB by activating the corresponding HWL, effectively eliminating the interference of other LCB cell values during the reading process. As shown in Fig. 6(a) and (b), compared with 6T-SRAM, RSNM shows a 1.3 times improvement in both 0.7-V low-voltage environment and 0.9-V normal voltage. From Fig. 6(c) and (d), it can be seen that the WSNM of 6T-DBCC slightly decreases, which is due to the loss caused by the transmission of GBL/GBLB voltage to BL/BLB through nMOS transistor during writing. However, its WSNM can still ensure circuit stability and normal function. Fig. 6(e) and (f) compares the simulation results of read and write delays between 6T-SRAM and 6T-DBCC under low/normal voltage, wide temperature range, and different corners. The results indicate that 6T-DBCC exhibits faster read and write speeds under any combination of voltage, temperature, and corner. This speed improvement primarily stems from the hierarchical bitline structure. While the access transistors (MN1/MN2) introduce additional series resistance, the dominant *RC* delay is reduced because the total capacitive load is significantly lowered. The GBL/GBLB has lower capacitance per unit length, and the local bitlines within each LCB are extremely short. Thus, converting the traditional large, concentrated bitline capacitance into a distributed combination of smaller capacitances results in a smaller overall *RC* time constant, as evidenced by the consistently lower delays in Fig. 6(e) and (f).

To comprehensively validate the storage stability against process variations and extreme operating conditions, extensive Monte Carlo (MC) simulations were performed across multiple critical PVT corners (e.g., SS/125 °C/0.7 V and FF/−40 °C/0.9 V). The statistical results in Fig. 6(g) confirm that the 6T-DBCC cell maintains sufficient and robust noise margins (RSNM >85 mV and WSNM >156 mV) with tight distributions under all tested scenarios, demonstrating reliable data read and write functionality.

B. Performance Simulation of Boolean Logic Operations

Taking the OR and AND operations on two operands 1 and 0 as an example, the simulation results are shown in Fig. 7(a) and (b). Based on the Boolean logic operation principle in Section II-B of the second section, it can be seen that in Fig. 7(a), NBL discharges $2\Delta V$, PBL discharges ΔV, and

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2000 Monte Carlo				RSNM		WSNM	
PVT	Corner	Temperature (°C)	Voltage (V)	Mean (mV)	Std (mV)	Mean (mV)	Std (mV)
WC	SS	125	0.7	85.03	7.91	164.99	21.2
BCF	FF	-40	0.9	118.36	9.48	196.43	22.64
ML	FF	125	0.9	89.17	7.65	211.73	19.13
WC-LT	SS	-40	0.9	110.49	8.43	194.94	25.57
BC-LV	FF	25	0.7	88.6	8.18	156.3	20.69
TL	TT	25	0.9	119.46	9.04	191.8	23.27

Fig. 6. Comparison of RSNM at (a) 0.7 V and (b) 0.9 V and WSNM at (c) 0.7 V and (d) 0.9 V between 6T and 6T-DBCC. Comparison of (e) read delay and (f) write delay between 6T and 6T-DBCC. (g) RSNM and WSNM of key PVT combinations WC timing, best-case fast (BCF), maximum leakage (ML), WC timing in low temperature (WC-LT), BCF in low voltage (BC-LV), and TL for 2000 MC simulation results of the 6T-DBCC.

ultimately VNBL < VPBL. SA outputs 1, and the OR result is 1 OR 0 = 1. In Fig. 7(b), NBL discharges ΔV, PBL discharges $2\Delta V$, and ultimately VNBL > VPBL. SA outputs 0, and the AND result is 1 AND 0 = 0. Therefore, the 6T-DBCC architecture can correctly perform Boolean logic operations.

This design uses the voltage-sensitive SA to directly detect the voltage difference between PBL and NBL, eliminating the need for V_{ref}, and the SA can recognize a minimum of 40 mV. We verify the compatibility of Boolean logic operations and SA through MC simulation using the 6T-DBCC architecture. Fig. 7(c) and (d) shows that under the conditions of 0.9 V, room temperature, and TT corner, after 2000 MC simulations, the mathematical expectation (Mean) ΔV values between PBL and NBL in a single OR/AND operation of 6T-DBCC were 87.82 and 82.23 mV, respectively, which were significantly higher than the identification value of SA 40 mV. Meanwhile, their standard deviations (Std) are 6.3 and 7.5 mV, ensuring that the Boolean operation results can be reliably recognized by the SA. Fig. 7(e) shows the energy simulation results of a single Boolean logic operation performed by 6T-DBCC at different temperatures and corners, with a power consumption range of 8.39–32.4 fJ, which is basically equivalent to the Boolean logic operation energy consumption reported in [23] (11.22–29.67 fJ), indicating that the Boolean operation energy consumption of this design is at a reasonable level. This design has a single Boolean logic operation with an energy consumption of only 11.29 fJ under TT corner and 27 °C room temperature conditions, fully meeting the low-power requirements of conventional operations.

To further evaluate the robustness of the Boolean logic scheme to inherent timing variations in the (N−1) ΔT offset method, extended simulations were conducted for Boolean logic computations involving three operands and four operands in Fig. 8. In Fig. 8(a), for three-operand cases (requiring a $2\Delta T$ bias pulse), the differential voltage ΔV for all eight input logic states ranges from 55.43 to 68.71 mV. In Fig. 8(b), for four-operand cases (requiring a $3\Delta T$ bias pulse), the differential voltage ΔV for all ten input states ranges from 54.38 to 70.60 mV. In all cases, the resulting ΔV significantly exceeds the SA's offset threshold of approximately 40 mV. These results confirm the circuit's functional correctness and demonstrate ample signal margin under the designed multioperand timing scheme, thereby substantiating its robustness against practical timing nonidealities.

C. Performance Simulation of MAC Operations

Stability is the core consideration for a circuit, to this end, we first conducted 2000 MC simulations of PBL/NBL discharge at $1\Delta V$ each under different corners, a low voltage of 0.7 V, and a normal voltage of 0.9 V at 27 °C, to examine the voltage changes of PBL/NBL and verify the stability of the 6T-DBCC structure, the results are reflected in Fig. 9(a)–(d). Overall, under the SS corner, regardless of low-voltage or normal voltage environments, the residual voltage Std of PBL and NBL after discharging ΔV is the smallest, and the Std is the best. In the same corner, the Std performance is better under a low-voltage environment. Comparing Fig. 9(a) and (b), as well as Fig. 9(c) and (d), the voltage mean difference between NBL and PBL varies significantly with voltages and corners. The difference is the smallest at 0.9 V and the TT corner, only |889.4–888.5 mV| = 0.9 mV, while the difference is the largest at 0.7 V and the FF corner, |682.8–691.1 mV| = 8.3 mV. Therefore, the 6T-DBCC architecture has the smallest discharge difference between PBL and NBL when operating at a normal voltage of 0.9 V and the TT corner, which can better meet the voltage synchronization accuracy requirements of

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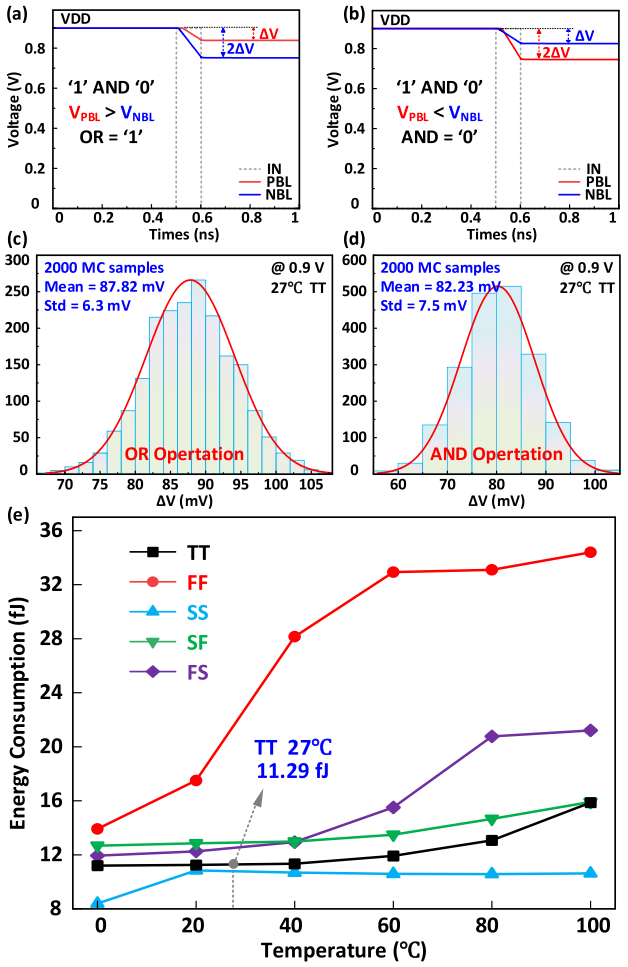


Fig. 7. Result of a single (a) OR or (b) AND operation with two operands. Analysis in 2000 MC simulation for single (c) OR or (d) AND operation with two operands. (e) Energy consumption simulation of single Boolean logic operations.

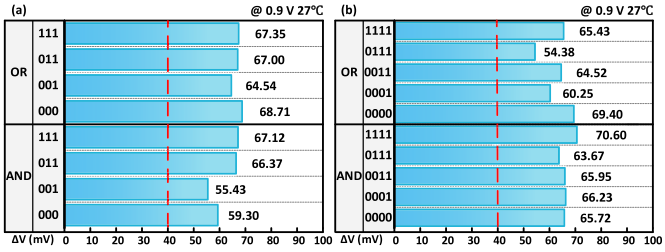


Fig. 8. Simulated ΔV for Boolean operations with (a) three and (b) four operands.

MAC operation. Similarly, under the same voltage and corner, the Std of the residual voltage after PBL discharge, ΔV, is slightly higher than that of NBL, indicating that the discharge stability of PBL is slightly lower than that of NBL. From the schematic, the differences in discharge between PBL and NBL can be analyzed. From the 6T-DBCC structure, it can be seen that the PBL discharge path has one more inverter delay than the NBL, resulting in the PBL discharge amount being smaller than the NBL discharge amount within the same time. To maintain the same discharge rate between the two, the design increased the NBL load capacitance to match the NBL discharge rate with the PBL discharge path, and further conducted MAC operation performance simulation.

The MAC operation achieves different calculation results by controlling the number of 6T-SRAM cells in the same row of LCB, accumulating analog voltages on the PBL/NBL bitlines. Its linearity is measured by the integral nonlinearity (INL), and the INL calculation formula is shown in (4) and (5). Among them, $V_{\max}$ is calculated for all 32 LCBs, and $V_{\min}$ is the discharge value calculated for 0 LCBs, $n = 32$, V_{cal} is the current discharge capacity value of PBL/NBL, and m is the current calculated number of LCBs

$$\Delta V = \frac{V_{\max} - V_{\min}}{n} \tag{4}$$

$$\text{INL} = \frac{V_{\text{cal}} - V_{\min}}{\Delta V} - m. \tag{5}$$

Fig. 9(e)–(h) shows the PBL/NBL discharge and INL results of 6T-DBCC during MAC operation at different voltages and corners. Combining Fig. 9(e) and (f), it can be observed that under TT corner and 27 °C environment when the same number of LCBs is activated to perform MAC operation at a low voltage of 0.7 V or a normal voltage of 0.9 V, the residual voltage $V_{\text{cal_NBL}}$ after NBL/PBL discharge remains highly consistent with $V_{\text{cal_PBL}}$, verifying the rate matching characteristics of the dual pull-down discharge path. According to the industry standard in [29], when the INL value is below 3 least significant bit (LSB), the circuit can be considered to have stable linearity, and the smaller the maximum INL value, the better the linearity and the higher the circuit reliability. Specific data show that at a voltage of 0.7 V, the maximum INL of NBL/PBL is 1.03 LSB and 1.02 LSB. At a voltage of 0.9 V, both decrease to 0.59 LSB. This result is not only significantly lower than the threshold requirement of 3 LSB but also confirms through comparison that the 6T-DBCC architecture can achieve optimal reliability at a working voltage of 0.9 V. Fig. 9(g) and (h) jointly indicates that when the 6T-DBCC architecture performs MAC operations in low-voltage environments, regardless of the process corner selected, its INL will significantly deteriorate. This is mainly due to the decrease in signal processing accuracy caused by insufficient power supply, which affects linearity. Specific data show that, except for the INL value of NBL/PBL exceeding 3 LSB under 0.7-V condition and FF corner condition, the INL values are significantly lower than 3 LSB under other operating conditions, indicating that the circuit reliability meets design expectations.

The proposed 6T-DBCC architecture adopts a dual pull-down discharge path to represent the positive and negative BWN MAC results, and improves circuit stability by enhancing the ability to resist process fluctuations. Compared with circuit designs that support similar functions, the comparison of process stability is shown in Fig. 10. To ensure the validity of the comparison, we conducted a ΔV simulation comparison in the 0.9 V, TT/SF/FS corner environment with the best LAU [19] circuit performance. The detailed data results are recorded in Fig. 10(a), indicating that temperature and corner changes have a significant impact on the discharge ΔV during MAC operation. At 27 °C, ΔV of LAU varies from 8.18 to 12.99 mV with a fluctuation amplitude of 4.81 mV as the

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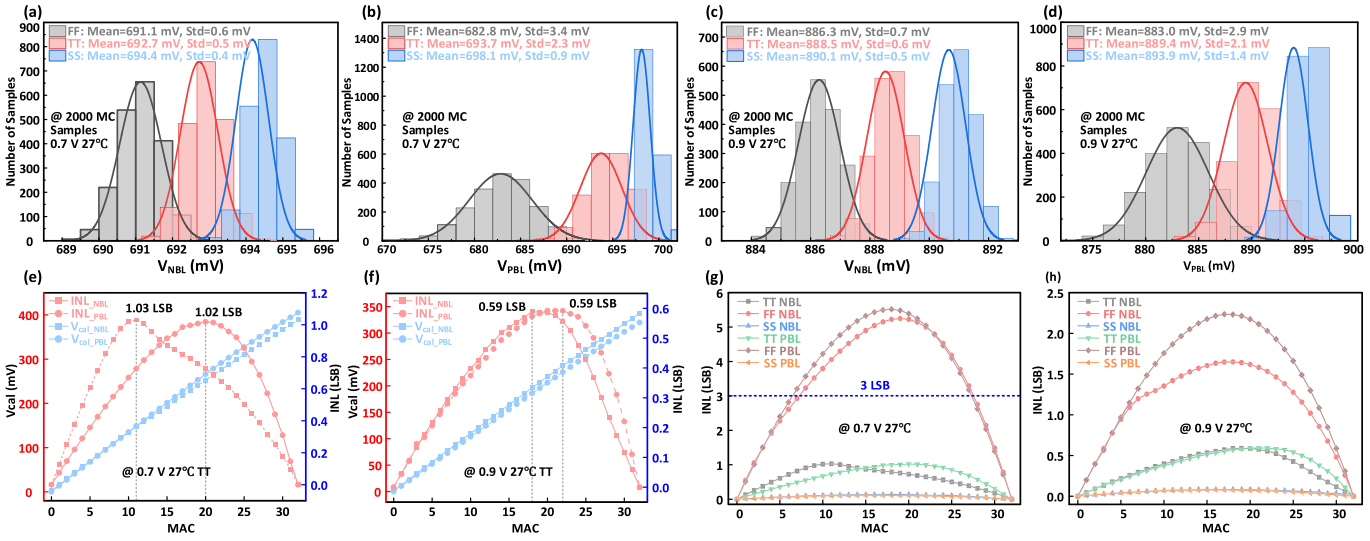


Fig. 9. 2000 MC simulations of MAC operation discharge ΔV at 0.7 V for (a) NBL or (b) PBL and 0.9 V for (c) NBL or (d) PBL. Reliability analysis of MAC operation of 6T-DBCC at (e) 0.7 V and (f) 0.9 V under TT corners, as well as comparison results of INL at different corners under (g) 0.7 V and (h) 0.9 V.

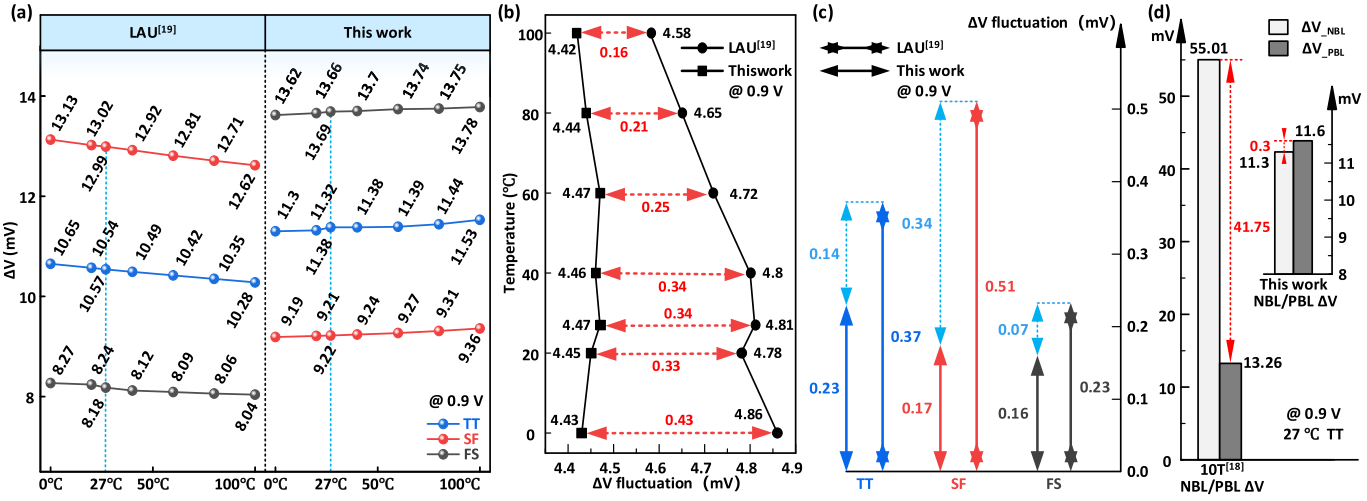


Fig. 10. 6T-DBCC performs MAC operations on (a) ΔV difference with LAU [19], (b) ΔV fluctuation with LAU [19] at the same temperature but different corners, (c) ΔV fluctuation with LAU [19] at the same corner but different temperatures, and (d) positive and negative consistency comparison with 10T [18].

corner changes, while ΔV of 6T-DBCC varies from 9.22 to 13.69 mV with a fluctuation amplitude of 4.47 mV, which is smaller than that of LAU and has a better performance. As shown in Fig. 10(b), further analysis of the fluctuation range of ΔV with the change of process corner under the same temperature conditions reveals that the ΔV fluctuation range of 6T-DBCC is significantly smaller than that of LAU at all temperatures. Especially at 0 °C, the ΔV fluctuation range of 6T-DBCC in the TT/SF/FS corners combination is 4.43 mV, which is 8.8% lower than LAU's 4.86 mV. In Fig. 10(c), under all corners, the fluctuation range of ΔV generated by 6T-DBCC with temperature changes is smaller than that of LAU. Especially under the SF corner, the ΔV fluctuation range of 6T-DBCC is only 0.17 mV, which is about 66.7% lower than LAU's 0.51 mV. From the above, it can be seen that the LAU achieves positive and negative results of MAC operation through the charging and discharging path. However, due to the poor process tolerance of the pull-up charging path, the ΔV generated by its charging and discharging is sensitive to temperature and process corner changes, making it difficult to maintain a stable ΔV value, resulting in a decrease in computational stability. The 6T-DBCC innovatively uses a dual pull-down circuit path to complete MAC symbol operation, which has a relatively stable ΔV value and can effectively improve system stability. Maintaining consistent positive and negative discharge rates is an important consideration for the accurate implementation of MAC operation in the dual pull-down discharge path. Therefore, we compared the 10T structure in Fig. 10(d) with that in [18], which uses positive and negative ΔV accumulation on the same path to achieve MAC operation. At the TT corner 27 °C and 0.9 V, ΔV of 6T-DBCC PBL is 11.3 mV, and ΔV of NBL is 11.6 mV, with a difference of only 0.3 mV between the two. However, the difference in ΔV between the positive and negative bitlines of

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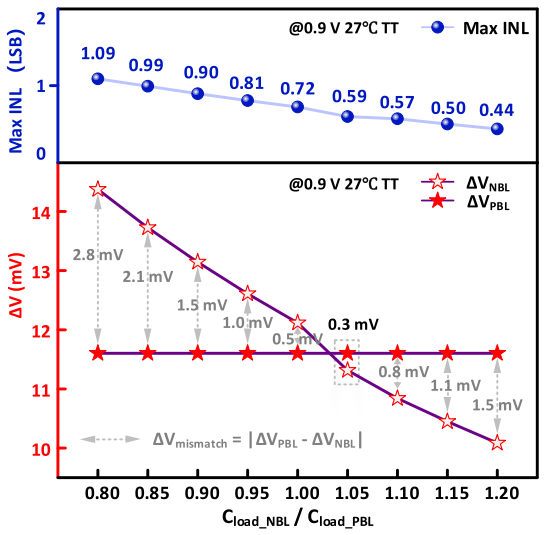


Fig. 11. Effect of NBL-to-PBL capacitance ratio on discharge matching and computational linearity.

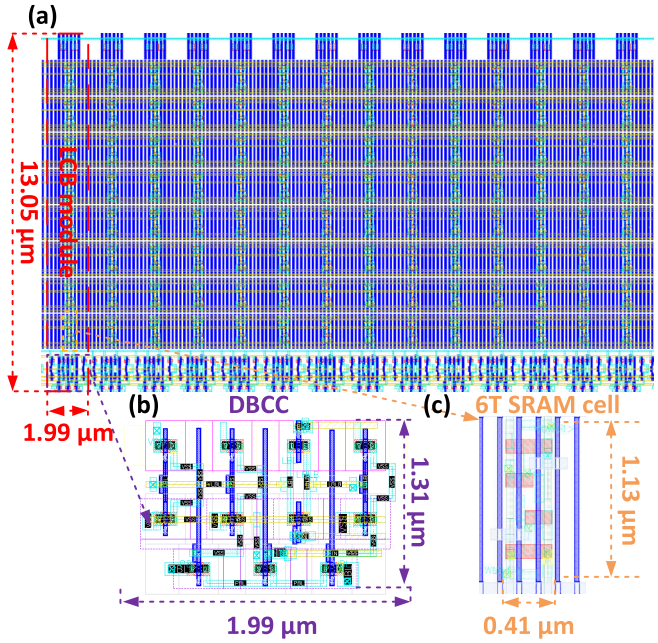


Fig. 12. Layout of (a) 6T-DBCC array, (b) DBCC, and (c) standard 6T-SRAM cell.

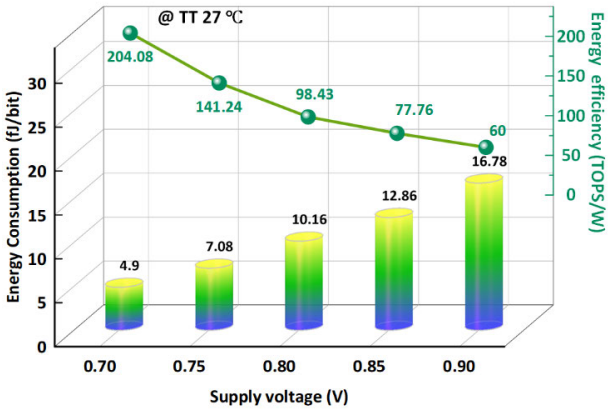


Fig. 13. Energy consumption and energy efficiency of different supply voltages in the 6T-DBCC architecture.

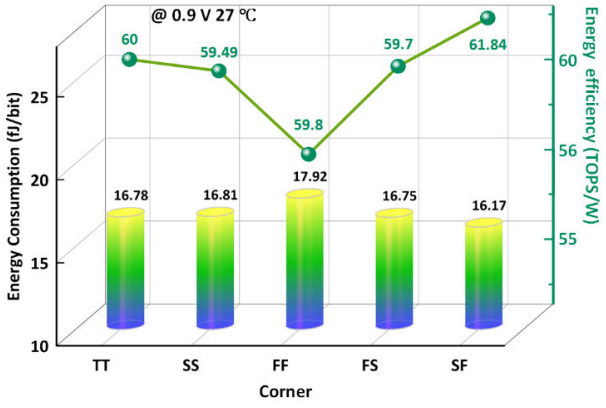


Fig. 14. Energy consumption and energy efficiency of different corners in the 6T-DBCC architecture.

the 10T structure is as high as 41.75 mV, which is significantly higher than that of the 6T-DBCC design, and the calculation accuracy is even worse.

This excellent path-matching performance (0.3-mV mismatch) results from a systematic optimization of the load capacitances on the dual pull-down paths. The buffer in the PBL path introduces a fixed delay, which we compensated for by finely tuning the load capacitance on NBL (Cload_NBL) to balance the discharge time constants of the two paths. A detailed parameter analysis in Fig. 11 shows that when the C_{load_NBL}-to-C_{load_PBL} ratio is approximately 1.05, the path voltage mismatch ($|\Delta V_{PBL} - \Delta V_{NBL}|$) reaches a submillivolt minimum, and the MAC INL concurrently approaches its optimum (<1 LSB). This passive, design-time tuning method fundamentally ensures the computational linearity and accuracy advantage of the dual-nMOS pull-down architecture.

The 6T-DBCC architecture layout implemented using TSMC 28-nm technology is shown in Fig. 12: (a) the 6T-DBCC array, (b) the DBCC, and (c) a standard 6T-SRAM cell. The innovative DBCC cell size is 1.99 × 1.31 μm, which includes both the storage and computing circuitry. MAC operation is the core operation of BWN, and its energy efficiency directly affects the application cost and range. Therefore, we use the average current method to quantify power consumption, taking the product of average current and voltage as instantaneous power, and then multiplying it by the MAC operation time to obtain single energy consumption. We measure computing power by the ratio of trillions of operations per second to power consumption. Fig. 13 shows the energy consumption and energy efficiency changes of the 6T-DBCC architecture when performing MAC operations with different supply voltages at the TT corner. When the operating voltage is set to 0.7 V, the system exhibits the optimal energy efficiency performance, with a per-bit energy consumption of only 4.9 fJ and an energy efficiency of 204.8 TOPS/W. As the supply voltage increases to 0.75 V, the system energy consumption increases by 44.5% to 7.08 fJ/bit, and the corresponding energy efficiency decreases to 141.24 TOPS/W. When the voltage continues to rise to 0.8 V, the energy consumption further increases to 10.16 fJ/bit, and the energy efficiency decreases

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TABLE V
PERFORMANCE COMPARISON TABLE

		VLSI'19[18]	VLSI'21[22]	JSSC'22[12]	TCAS-II'23[19]	JSSC'24[13]	This work
Technology (nm)		28	28	65	28	28	28
Supply voltage (V)		1	0.5-0.9	0.5-1.0	0.9	0.9	0.7-0.9
Array size		256 × 64	256 × 64	256 × 64	256 × 64	256 × 64	256 × 64
Memory capacity		16 Kb	16 Kb	16 Kb	32 Kb	16 Kb	16 Kb
Type of supported function		Memory/ MAC	Memory/ Logic/MAC	Memory/ MAC	Memory/ MAC	Memory/ MAC	Memory/ Logic/MAC
Computation mode		Analog	Digital	Analog	Analog	Analog	Analog
Input precision		1	1	1	1/2/4	4/8	5
Weight precision		1	1	1	1/2/4	8	1
Energy efficiency (TOPS/W)		≈ 300[a]	≈ 175.4[b] (0.5 V)	15.8-490[a]	47-155.6[b]	22.2[a]	204.8[b] (0.7 V) 55.8[b] (0.9 V)
Throughput (GOPS)		NA[c]	NA[c]	NA[c]	256[b]	51.2[a]	64[b]
Power (fj/bit)		3.33[a]	5.7[b] (0.5 V) 7.8[b] (0.6 V)	NA[c]	NA[c]	NA[c]	4.9[b] (0.7 V) 16.78[b] (0.9 V)
Area efficiency (TOPS/mm^2)		NA[c]	NA[c]	112[a]	5.03[b]	1.6[a]	1.2[b]
Operating frequency		25 MHz[a]	2.44 GHz[b]	NA[c]	NA[c]	≈ 200 MHz[a]	250 MHz[b]
Accuracy	**MNIST**	98.42%[a]	NA[c]	96.37%[a] 97.04%[b]	NA[c]	NA[c]	97.59%[b]
	CIFAR-10	79.38%[a]	NA[c]	87.1%[a] 88.14%[b]	85.2-85.7%[b]	90.92[a] 91.26%[b]	89.08%[b]

a. Based on test results; b. Based on simulation results; c. No back calculation possible due to missing date.

to 98.43 TOPS/W. At a working voltage of 0.9 V, the system energy consumption reached 16.78 fJ/bit, and the energy efficiency decreased to 60 TOPS/W. These data clearly indicate a positive correlation between energy consumption and supply voltage, while energy efficiency has a negative correlation with supply voltage. Fig. 14 further analyzes the performance of energy consumption and energy efficiency under different corners at a voltage of 0.9 V, where the system energy consumption at the SS corner is 16.81 fJ/bit and the energy efficiency is 59.49 TOPS/W. The FF corner exhibits the worst performance, with an energy efficiency of only 55.8 TOPS/W and a maximum energy consumption of 17.92 fJ/bit. The energy consumption under the FS corner is 16.75 fJ/bit, and the energy efficiency is 59.7 TOPS/W. The comprehensive performance of the system is most balanced under the SF corner, with the lowest energy consumption of 16.17 fJ/bit and the highest energy efficiency of 61.84 TOPS/W. The experimental results confirm the significant impact of corners parameters on circuit performance.

Table V compares and analyzes this work with existing related studies and shows certain advantages in energy efficiency. Its energy efficiency reaches 204.8 TOPS/W at a supply voltage of 0.7 V, which is not only higher than the energy efficiency performance of [22] at about 175.4 TOPS/W at 0.5 V but also better than the highest energy efficiency of [19] at 155.6 TOPS/W and [13] at 22.2 TOPS/W. At the same time, compared with [12] energy efficiency range of 15.8–490 TOPS/W, the energy efficiency of this work at this voltage is at a relatively high level. Even at a supply voltage of 0.9 V, the energy efficiency of this work is still 55.8 TOPS/W, which is also higher than [12] 22.2 TOPS/W, demonstrating excellent energy efficiency characteristics under different voltage conditions. Furthermore, the design also attains a throughput of 64 GOPS, an operating frequency of 250 MHz, and an area efficiency of 1.2 TOPS/mm^2, as summarized in Table V. These metrics, achieved while integrating storage, multioperand Boolean logic, and signed MAC operations within a unified DBCC architecture, reflect the design tradeoff for high functional density. Overall, this work has a competitive advantage in energy efficiency compared to related research.

Furthermore, to assess the practical impact of circuit-level variations on application performance, a system-level evaluation was conducted. A circuit-algorithm

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cosimulation framework was established, injecting worst case (WC) hardware nonidealities (e.g., INL and mismatch) extracted from PVT simulations into the inference process of binary weight networks on the MNIST and CIFAR-10 datasets. As summarized in Table V, the design maintains high inference accuracies of 97.59% on MNIST and 89.08% on CIFAR-10, with a mere 0.74% accuracy degradation on CIFAR-10 compared to the ideal baseline. This result indicates that the circuit's inherent robustness, afforded by architectural features such as differential sensing and symmetric computation paths, effectively contains PVT-induced variations, thereby ensuring reliable application-level performance without necessitating additional on-chip compensation mechanisms.

V. Conclusion

This letter presents a 16-kb SRAM-based CIM macro utilizing a novel DBCC architecture designed and simulated in 28-nm CMOS technology. The proposed design successfully integrates three operational modes: conventional memory access, multioperand Boolean logic operations, and BWN MAC computations—within a unified memory array, addressing several critical challenges in conventional CIM designs.

First, the DBCC structure enables the storage of both positive and negative weights within a single 6T-SRAM cell, effectively doubling the array utilization compared to methods that require separate cells for different polarities. This not only reduces area overhead but also enhances inference speed and energy efficiency. Second, the symmetric pull-down discharge paths for both PBL and NBL ensure consistent discharge rates for positive and negative computations, significantly improving linearity and tolerance to process variations. MC simulations confirmed that INL remains below 3 LSB across various corners, at 0.9-V supply voltage. Third, the differential sensing mechanism for Boolean logic operations eliminates the need for an external V_{ref}, reducing peripheral circuit complexity and enhancing robustness against PVT variations.

The proposed macro also incorporates a compact 2:1 multiplexer-based PWM circuit for 5-b signed input encoding, which minimizes area overhead while maintaining functional flexibility. Energy efficiency evaluations demonstrated a range of 55.8–204.8 TOPS/W, depending on supply voltage and process corner, with optimal performance achieved at 0.7 V. These results compare favorably with state-of-the-art designs, as summarized in Table V, highlighting the competitive advantage of the proposed architecture in terms of energy efficiency, functional integration, and process robustness.

Furthermore, the design exhibits strong resilience to temperature and process corner variations, maintaining stable operation across a wide range of conditions. The dual discharge path structure not only improves computational accuracy but also reduces sensitivity to power supply noise and parasitic effects, making it suitable for mobile and edge-computing applications where power and stability are critical. System-level evaluations on MNIST and CIFAR-10 datasets further affirm its practicality, yielding competitive accuracies of 97.59% and 89.08%, respectively.

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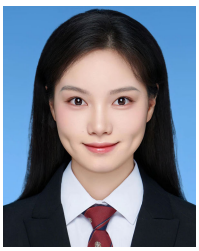
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